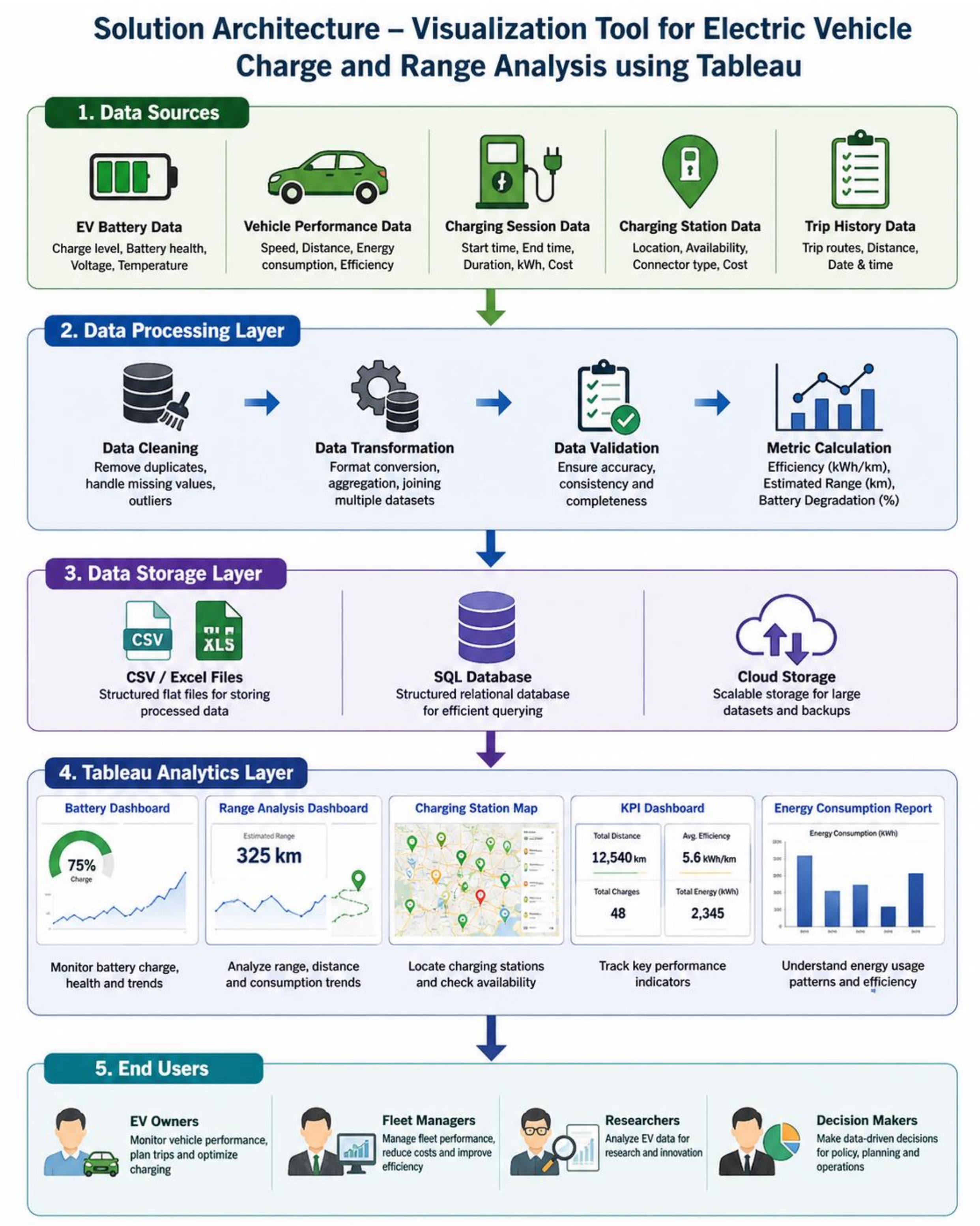
TEAM ID	SWTID-2026-6767
PROJECT TITLE	Visualization Tool for Electric Vehicle Charge and Range Analysis using Tableau
MAX MARKS	5 MARKS

4.3 – Solution Architecture

This solution architecture. Analyzes Electric Vehicle charging and range data from sources like EV sensors, Battery Management Systems, charging stations, GPS devices and weather services. Data is gathered through APIs or streaming platforms. It is then processed using ETL workflows. Stored in a centralized data warehouse. The processed data generates metrics such as battery health, energy consumption and charging efficiency. It also predicts the driving range of Electric Vehicles.

We use Tableau dashboards to visualize these insights. This enables users to monitor charging patterns, battery performance, costs and fleet efficiency. Security features like role-based access control and data governance ensure data access. The solution helps optimize Electric Vehicle Operations It improves battery performance reduces costs and enhances fleet management. We get Electric Vehicle performance and lower costs. This solution makes Electric Vehicle fleet management more efficient. It provides insights, into Electric Vehicle charging and range.


Solution Architecture – Visualization Tool for Electric Vehicle Charge and Range Analysis using Tableau
1. Data Sources
EV Battery Data
Charge level, Battery health, Voltage, Temperature
Vehicle Performance Data
Speed, Distance, Energy consumption, Efficiency
Charging Session Data
Start time, End time, Duration, kWh, Cost
Charging Station Data
Location, Availability, Connector type, Cost
Trip History Data
Trip routes, Distance, Date & time
2. Data Processing Layer
Data Cleaning
Remove duplicates, handle missing values, outliers
Data Transformation
Format conversion, aggregation, joining multiple datasets
Data Validation
Ensure accuracy, consistency and completeness
Metric Calculation
Efficiency (kWh/km), Estimated Range (km), Battery Degradation (%)
3. Data Storage Layer
CSV
XLS
CSV / Excel Files
Structured flat files for storing processed data
SQL Database
Structured relational database for efficient querying
Cloud Storage
Scalable storage for large datasets and backups
4. Tableau Analytics Layer
Battery Dashboard
75%
Charge
Range Analysis Dashboard
Estimated Range
325 km
Charging Station Map
KPI Dashboard
Total Distance
12,540 km
Avg. Efficiency
5.6 kWh/km
Total Charges
48
Total Energy (kWh)
2,345
Energy Consumption Report
Energy Consumption (kWh)
Monitor battery charge, health and trends
Analyze range, distance and consumption trends
Locate charging stations and check availability
Track key performance indicators
Understand energy usage patterns and efficiency
5. End Users
EV Owners
Monitor vehicle performance, plan trips and optimize charging
Fleet Managers
Manage fleet performance, reduce costs and improve efficiency
Researchers
Analyze EV data for research and innovation
Decision Makers
Make data-driven decisions for policy, planning and operations